Example 1. Let n be a positive integer. Prove that

$1 \leqslant \tau(n) \leqslant 2\sqrt{n}$

$n \leqslant \sigma(n) \leqslant 2n^{3/2}$

Proof. The lower bounds are easier to see.

The positive integer n has at least one divisor, namely n itself, and thus

Now let's address the upper bounds, beginning with a lemma.

Lemma. At least half of the positive divisors of n are less or equal $\sqrt{n}$.

Proof. Positive divisors of n can be uniquely paired—except for $\sqrt{n}$ itself—by

It can help to model this pairing with the divisors of 36 as below.

$1, 2, 3, 4, 6, 9, 12, 18, 36$

Each pair (or singleton, for $\sqrt{n}$ itself) contains a positive divisor that is:

Therefore, at least half of the positive divisors of n are less or equal $\sqrt{n}$.

□

Hence

$\frac{\tau(n)}{2} \leqslant$

Next, using the upper bound we have already proven for τ, we continue.

$\sigma(n) =$

□

A. **Convolutions.** There is a productive way to combine arithmetic functions.

Let f, g be arithmetic functions.

Their **Dirichlet convolution** is the arithmetic function $f * g$ with formula

$(f * g)(n) =$

For example

$(f * g)(6) =$

which suggests we can provide the more symmetric formula

$(f * g)(n) =$

where the summation notation should be read as

Note that the $*$ here should be read as "convoluted with" and not "multiplied with".

For example

$(f * \mathbb{1})(n) =$

And thus we may write

$\tau(n) = \sum_{d|n} 1 =$

$\sigma(n) = \sum_{d|n} d =$

Or put more succinctly

$\tau =$

$\sigma =$

B. **Properties of Convolutions.** Here we discuss properties of convolutions.

"Commutativity" is granted by the symmetric defintion of convolutions.

Let f and g be arithmetic functions. Then:

The next property is deemed "associativity" of convolutions.

Let f, g, h be arithmetic functions. Then:

The proof, which we shall not spell out in detail, boils down to showing that triple convolutions, independent of the placement of the parantheses may be written as

$(f * g * h)(n) =$

So, for example

$(f * g * h)(5) =$

Lastly we show that convolutions preserve multiplicativeness.

Let f and g be multiplicative. Then $f * g$ is also multiplicative.

Proof. First we use that $f(1) = 1$ and $g(1) = 1$ to conclude

$(f * g)(1) =$

Now suppose that $m \perp n$ are positive integers.

Because $m \perp n$, the divisors d of mn may be uniquely presented in the form

Then

$(f * g)(mn) =$

and then use multiplicativity of f and g to continue

$\cdots =$

and lastly use distributivity to conclude

$\cdots =$

C. **Unit Function.** The simple observation, which we already established, that

$\mathbb{1} * \mathbb{1} =$

shows that $\mathbb{1}$ has an effect when "convoluted" with other functions.

There is a function that has no effect when convoluted with other functions.

The **unit function** ϵ is the arithmetic function with formula

$\epsilon(n) =$ {

Let's show that ϵ is multiplicative.

ϵ is completely multiplicative.

Proof. Evidently $\epsilon(1) = 1$.

Next let m and n be positive integers greater than 1. Then

$\epsilon(mn) =$

$\epsilon(m)\epsilon(n) =$

□

Let's justify that ϵ has no effect in the world of convolutions.

This is what we would call the "unit" property.

Theorem. Let f be arithmetic. Then:

Proof. Let positive integer n be given. Then

$(f * \epsilon)(n) =$

and then because ϵ vanishes at every term of the form n/d except when:

$\cdots =$

□